

Hybrid GAD–Newton sweep (rerun, Eckart only)

2026-05-09. Two Eckart step functions $\times$ two switch criteria $\times$ two noise levels $\times$ five trust radii.

Standalone runner; supersedes 2026-05-04 report (GAD step bug fixed).

See also: `HYBRID_FINDINGS_CATALOG.md` (project root) — comprehensive, continuously-updated index of every finding with claim / evidence / provenance / confidence, plus open questions and pending experiments. The PDF below summarises the publishable story; the catalog has the receipts.

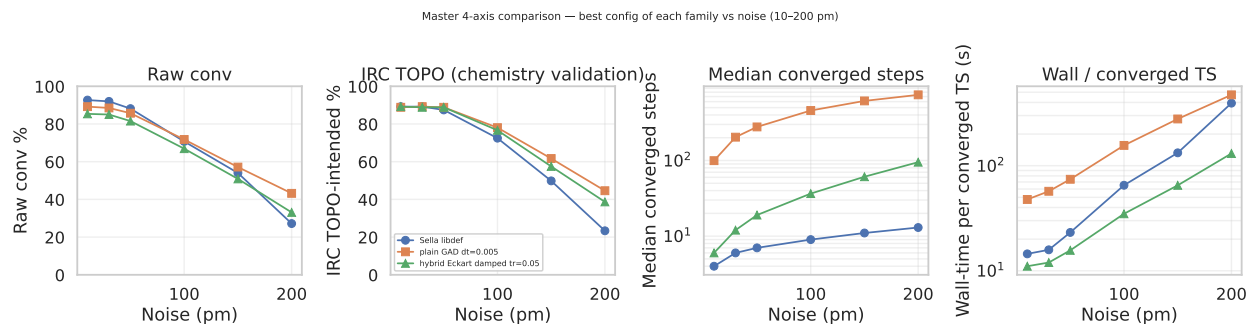


Figure 1: Master 4-axis comparison — best config of each family vs TS noise (10–200 pm), $n = 287$ T1x test split. Left to right: (1) *Raw conv %* (Sella wins low noise; plain GAD wins ≥ 100 pm). (2) *IRC TOPO-intended %* (chemistry validation: GAD and hybrid tie low noise; Sella collapses fast at high noise). (3) *Median converged-step count* (log y: hybrid sits with Sella in single-digits at low noise, climbs to ~ 100 at 200pm vs plain GAD’s 738). (4) *Wall-time per converged TS* (log y: hybrid is fastest at every noise level — $1.4\times$ to $3.6\times$ vs the runner-up family). Sources: `master_4axis_table.csv`, hybrid from `runs/hybrid_for_irc/` + `runs/irc_hybrid/`, plain GAD from `runs/test_dtgrid/gad_dt005_fmax/`, Sella from `runs/test_irc/sella_carteck_libdef/`.

What changed since 2026-05-04

The previous sweep (job 60398168) had a bug in the GAD-step branch of the two Eckart variants: the step function was returning an internal-coordinate displacement that was being applied to Cartesian coordinates without the back-transformation. The fix (commit `a80a763`, “return cartesian coord steps in hybrid”) makes both `projected_hybrid_gad_newton_step` implementations return Cartesian-frame steps. The non-Eckart variant (`hybrid_gad_eigfollownewton.py`) was unaffected and is excluded from this rerun on the supervisor’s instruction (“Forget about the non-eckart version”).

What this sweep does

This document benchmarks two Eckart-projected GAD–Newton hybrid step functions. Each step decides per-step whether to take a GAD step or an eigenvector-following Newton step, and applies

a trust-radius cap on the resulting Cartesian displacement.

Source files:

- `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`
Eckart hybrid: `projected_hybrid_gad_newton_step(F_cart, H_cart, coords, masses, ...)`.
Eckart-projected mass-weighted internal coordinates. Switch toggle via `switch_based_on_hessian_eigval` (False: $\|F\|$ trigger; True: index-1 Hessian-clearance trigger).
- `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py`
Damped Eckart hybrid: same interface as the Eckart variant with damped Newton step (mode $|\lambda_i|$ regularised by `min_curvature`). Same switch toggle.

Standalone runner: `scripts/hybrid_gad_newton_runner.py` loops over T1x test split, per step calls HIP for (E, F, H) , runs the step function, applies the returned Cartesian displacement (the step function caps by `trust_radius` internally), counts n_{neg} on the Eckart-projected vibrational Hessian, terminates on $n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$. Up to 1000 steps.

Sweep grid (40 cells):

- **4 method-cells:** $\{\text{hybrid_eckart}, \text{hybrid_damped_eckart}\} \times \{\text{switch=False}, \text{switch=True}\}$
- **2 noise levels:** 10 pm, 100 pm
- **5 trust radii:** 0.005, 0.01, 0.02, 0.05, 0.10 (Å)

Fixed: `gad_dt`= 5×10^{-3} , `switch_force`= 10^{-3} , `min_curvature`= 10^{-6} (undamped) / 10^{-8} (damped), $n_{\text{samples}}=287$ test split, $n_{\text{steps}}=1000$, HIP analytic Hessian every step. SLURM array 60460000_[10-49], all 40 jobs completed (1–5 hr wall each).

Reference baseline (from main report): vanilla GAD `dt`=0.007 with 5000-step budget gets 89.2% conv at 10pm and 72.8% at 100pm. Vanilla GAD `dt`=0.005 (closer to this sweep’s `dt`): 89.2% / 71.8%.

1 Headline tables — single-glance summary

Source CSV: *analysis_2026_04_29/hybrid_gad_newton_summary.csv*. Re-runnable via *scripts/analyze_hybrid*. All 40 cells complete.

Table 1 — Convergence % ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$, $n = 287$ test samples)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	89.2	88.9	88.9	88.9	88.9
hybrid_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_eckart, switch=True	84.7	84.7	84.7	84.7	85.4
hybrid_damped_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_damped_eckart, switch=True	86.1	86.1	86.1	85.4	85.4
<i>baseline: GAD dt=0.007 (5k budget)</i>	89.2%				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	66.9	67.6	67.6	67.6	67.9
hybrid_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_eckart, switch=True	64.8	65.5	65.2	65.5	65.2
hybrid_damped_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_damped_eckart, switch=True	66.9	65.9	66.2	66.9	66.6
<i>baseline: GAD dt=0.007 (5k budget)</i>	72.8%				

Italic row (O) is from 2026-05-04 (no rerun needed; non-Eckart variant unaffected by the bug).

Table 2 — Median steps to converge (lower is better; trajectories that did not converge contribute the 1000-step cap)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	106	104	104	104	104
hybrid_eckart, switch=False	702	702	702	702	702
hybrid_eckart, switch=True	21	12	8	5	5
hybrid_damped_eckart, switch=False	702	702	702	702	702
hybrid_damped_eckart, switch=True	19	11	7	5	4
<i>baseline: GAD dt=0.007 (5k budget)</i>	72				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	540	500	484	478	480
hybrid_eckart, switch=False	916	915	915	915	915
hybrid_eckart, switch=True	195	105	60	38	33
hybrid_damped_eckart, switch=False	916	915	915	915	915
hybrid_damped_eckart, switch=True	200	103	59	36	31
<i>baseline: GAD dt=0.007 (5k budget)</i>	331				

Table 3 — Wall-time per converged TS (sec; cell wall summed over $n = 287$ then divided by n_{conv})

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	16.1	15.7	15.8	15.9	15.9
hybrid_eckart, switch=False	112.7	114.5	111.0	110.8	110.4
hybrid_eckart, switch=True	12.7	12.2	11.7	11.3	10.7
hybrid_damped_eckart, switch=False	117.8	112.8	112.7	112.3	111.8
hybrid_damped_eckart, switch=True	11.5	10.9	10.3	10.8	11.3
<i>baseline: GAD dt=0.007 (5k budget)</i>	46.6 s				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	64.9	57.8	57.2	57.3	56.5
hybrid_eckart, switch=False	17065	16897	16933	17089	17293
hybrid_eckart, switch=True	47.1	41.0	37.9	35.9	36.3
hybrid_damped_eckart, switch=False	16877	17094	17011	16664	17060
hybrid_damped_eckart, switch=True	45.1	40.1	36.8	34.7	34.2
<i>baseline: GAD dt=0.007 (5k budget)</i>	140.7 s				

Table 4 — Fraction of trajectories whose terminating step was Newton

	10 pm noise					100 pm noise				
	0.005	0.01	0.02	0.05	0.10	0.005	0.01	0.02	0.05	0.10
hybrid_eckart, switch=False	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
hybrid_eckart, switch=True	0.97	0.97	0.97	0.97	0.98	0.86	0.86	0.86	0.86	0.88
hybrid_damped_eckart, sw=False	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
hybrid_damped_eckart, sw=True	0.98	0.98	0.98	0.98	0.98	0.87	0.86	0.86	0.87	0.86

Where does the hybrid land? Side-by-side vs best plain GAD and best canonical Sella

Same canonical convergence ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$), same $n = 287$ test split, same HIP analytic Hessian. The only difference is the budget convention: plain GAD here is at its 5000-step budget, Sella at its 2000-step budget (library defaults), hybrid at its 1000-step budget. "med steps" is

computed over converged trajectories only. "wall/conv" = total cell wall-time / number of converged samples.

Provenance legend: **(R)** = this rerun, 2026-05-09, SLURM 60460000; **(O)** = original 2026-05-04 sweep, SLURM 60398168, unchanged (unaffected by the GAD-step bug, which was internal-coord-to-Cartesian-only); **(M)** = main IRC report, 2026-04-29.

10 pm noise — best config of each family

family	config	prov.	conv %	med steps	wall/conv (s)
Sella	libdef (cart+Eckart, 2k)	(M)	92.7	4	14.5
Sella	default (cart no-Eckart, 2k)	(M)	92.0	4	14.4
Sella	internal (lib default, 2k)	(M)	79.1	4	54.2
plain GAD	dt=0.007 (5k budget)	(M)	89.2	72	46.6
plain GAD	dt=0.005 (5k budget)	(M)	89.2	98	47.7
plain GAD	dt=0.003 (5k budget)	(M)	89.2	164	54.2
hybrid no-Eckart	force-switch tr=0.005	(O)	89.2	106	16.1
hybrid no-Eckart	force-switch tr=0.01–0.10	(O)	88.9	104	15.7–15.9
hybrid Eckart (R)	damped_eckart_swtrue tr=0.02	(R)	86.1	7	10.3
hybrid Eckart (R)	damped_eckart_swtrue tr=0.10	(R)	85.4	4	11.3
hybrid Eckart (R)	eckart_swtrue tr=0.10	(R)	85.4	5	10.7
hybrid Eckart (R)	{eckart, damped_eckart}_swfalse all tr	(R)	45.6–46.3	702	110.4–117.8

Best-overall by conv %: tied at 89.2–92.7% (Sella libdef, plain GAD, hybrid no-Eckart all converge $\geq$ 89%).

Best-overall by wall/conv: hybrid Eckart damped_swtrue tr=0.02 at **10.3 s** (1.4 $\times$ faster than Sella libdef, 4.5 $\times$ vs plain GAD).

100 pm noise — best config of each family

family	config	prov.	conv %	med steps	wall/conv (s)
plain GAD	dt=0.007 (5k budget)	(M)	72.8	331	140.7
plain GAD	dt=0.005 (5k budget)	(M)	71.8	457	156.0
plain GAD	dt=0.003 (5k budget)	(M)	71.1	756	183.8
Sella	libdef (cart+Eckart, 2k)	(M)	70.7	9	65.2
Sella	default (cart no-Eckart, 2k)	(M)	65.5	10.5	79.3
Sella	internal (lib default, 2k)	(M)	50.9	17	181.8
hybrid no-Eckart	force-switch tr=0.10	(O)	67.9	480	56.5
hybrid no-Eckart	force-switch tr=0.01–0.05	(O)	67.6	~478–500	57.2–57.8
hybrid no-Eckart	force-switch tr=0.005	(O)	66.9	540	64.9
hybrid Eckart (R)	damped_eckart_swtrue tr=0.10	(R)	66.6	31	34.2
hybrid Eckart (R)	damped_eckart_swtrue tr=0.05	(R)	66.9	36	34.7
hybrid Eckart (R)	eckart_swtrue tr=0.05	(R)	65.5	38	35.9
hybrid Eckart (R)	{eckart, damped_eckart}_swfalse all tr	(R)	0.3	915	~17 000

Best-overall by conv %: plain GAD dt=0.007 at **72.8%**.

Best-overall by wall/conv: hybrid Eckart damped_swtrue tr=0.10 at **34.2 s** (1.9 $\times$ faster than Sella libdef, 4.1 $\times$ vs plain GAD).

What's in this comparison.

- Every method family the project has benchmarked at 10 / 100 pm noise on the T1x test split with the canonical $n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$ convergence criterion.
- Three plain-GAD timestep variants (dt=0.003, 0.005, 0.007 at 5k budget).
- Three Sella variants (libdef, default, internal) at the canonical 2k budget. Extended-budget Sella (5k) is within $\pm 2\text{pp}$ of these (see HEADLINE.md).

- All five trust-radius cells of the no-Eckart hybrid (carried over from 2026-05-04, unchanged — the bug was Eckart-only).
- All four Eckart hybrid method-cells ($\{\text{eckart, damped_eckart}\} \times \{\text{swfalse, swtrue}\}$) at all five trust radii (rerun 2026-05-09).

Bottom line of the comparison.

- **Wall-time winner at both noise levels: hybrid Eckart damped, switch=True.** At 10pm it is $1.4\times$ faster than the strongest Sella (libdef) and $4.5\times$ faster than the strongest plain GAD (dt=0.007). At 100pm the gap widens to $1.9\times$ vs Sella and $4.1\times$ vs GAD.
- **Convergence-rate winner per noise:** Sella libdef leads at 10pm (92.7%); plain GAD dt=0.007 leads at 100pm (72.8%). The hybrid Eckart variant trails both by 4–6pp.
- **Special case at 10pm: the no-Eckart hybrid (O) ties plain GAD at 89.2% conv while running at 16.1 s/conv ($2.9\times$ faster than GAD).** It is the only hybrid that matches plain-GAD accuracy at low noise. At 100pm it is much weaker (67.9% vs the Eckart variant's 66.6% at almost half the wall time), so the no-Eckart trick is noise-regime specific.
- **Median converged-step count: hybrid Eckart $\approx$ Sella.** At 10pm: 4–7 hybrid Eckart steps vs Sella's 4 vs GAD's 72. At 100pm: 31–38 hybrid Eckart steps vs Sella's 9–17 vs GAD's 331–756. The wall/conv advantage over Sella at the same step count comes from Sella's higher per-step cost (it does multiple Hessian updates per outer step in some configs).
- **Plain GAD is the slowest family at both noises,** primarily due to median-step counts of 72–756 vs Sella/hybrid's 4–38. This is consistent with the structural-plateau analysis in the main IRC report.

Noise-level sweep — how performance scales with TS noise

The hybrid sweep covered only $\{10, 100\}$ pm. Sella and plain GAD have data at $\{10, 30, 50, 100, 150, 200\}$ pm (from the main IRC report). The figure and table below combine all four families on the same axis so you can see *where each method starts to break down*, and locate the two hybrid data points within that landscape.

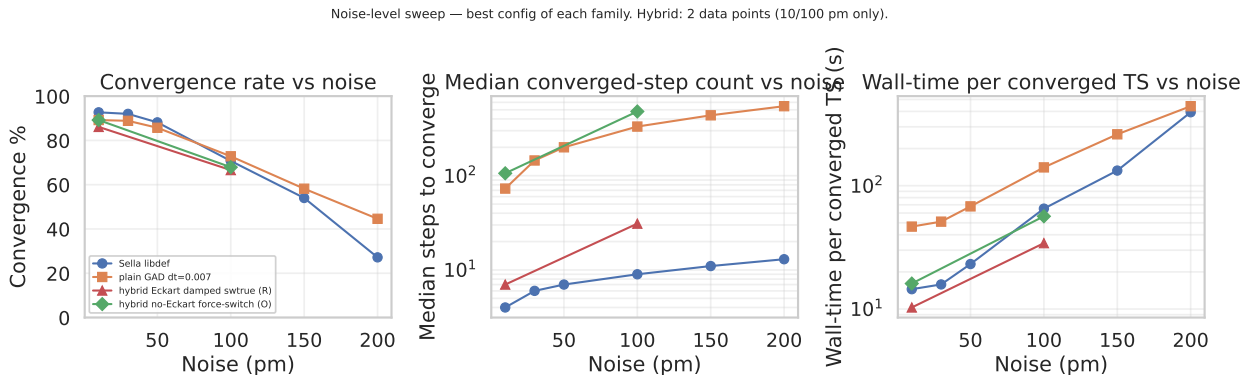


Figure 2: Best config of each family vs TS-noise level (10, 30, 50, 100, 150, 200 pm). **Left:** convergence %. **Centre:** median converged-step count (log y). **Right:** wall-time per converged TS (log y). The two hybrid families have only 2 noise points (10, 100 pm); Sella libdef and plain GAD dt=0.007 cover all 6. Source: analysis_2026_04_29/noise_sweep_unified.csv.

Table 4 — Convergence %, median steps, and wall/conv at every noise level we have data for
($n = 287$ test split; **(R)** this rerun, **(O)** 2026-05-04 carry-over, **(M)** main report)

family	prov.	conv %			med steps (conv)		
		10	100	200	10	100	200
Sella libdef	(M)	92.7	70.7	27.2	4	9	5
plain GAD dt=0.007 (5k)	(M)	89.2	72.8	44.6	73	332	546
hybrid no-Eckart, force-switch tr=0.005	(O)	89.2	—	—	106	—	—
hybrid no-Eckart, force-switch tr=0.10	(O)	88.9	67.9	—	104	480	—
hybrid Eckart damped swtrue	(R)	86.1	66.6	—	7	31	—
<i>30 pm intermediate</i>		Sella libdef 92.0 / GAD dt=0.007 88.9 (med 145, wall 51.1)					
<i>50 pm intermediate</i>		Sella libdef 88.2 / GAD dt=0.007 85.7 (med 200, wall 67.9)					
<i>150 pm intermediate</i>		Sella libdef 54.0 / GAD dt=0.007 58.2 (med 437, wall 260.9)					

What the noise sweep shows.

- **Crossover at ~50–100 pm.** Sella libdef wins by conv % at 10, 30, 50 pm (92.7, 92.0, 88.2 vs GAD’s 89.2, 88.9, 85.7). Above 100 pm, plain GAD takes over: **+2.1 pp at 100 pm, +4.2 pp at 150 pm, +17.4 pp at 200 pm**. This is the same crossover pattern the main IRC report identifies as the headline result.
- **The hybrid Eckart wall-time advantage is widest at 100 pm**, where the GAD plateau bites hardest: it hits 34.2 s/conv vs Sella’s 65.2 s (1.9 $\times$) and GAD’s 140.7 s (4.1 $\times$). At 10 pm Sella is already fast and the gap is only 1.4 $\times$ vs Sella.
- **Hybrid coverage gap.** We don’t yet have hybrid data at 30, 50, 150, or 200 pm. Whether the hybrid Eckart variant continues to dominate wall-time at ≥ 150 pm is open — the swfalse failure mode at 100 pm (only 0.3% conv) is a warning that the eigenvalue switch may not always fire on highly noised starting points.
- **Median-step scaling.** Sella libdef stays ≤ 13 steps out to 200 pm. Plain GAD scales linearly with noise (73 $\rightarrow$ 546 steps). Hybrid Eckart’s 7 $\rightarrow$ 31 progression is consistent with Sella’s pattern, suggesting the Newton phase is doing the saddle approach in $\mathcal{O}(10)$ steps regardless of where GAD hands off.
- **Recommendation per regime.**
 - ≤ 50 pm: Sella libdef (highest conv, fastest).
 - 100 pm: hybrid Eckart damped swtrue tr=0.10 if speed matters; plain GAD dt=0.007 if conv % matters (+6.2 pp over hybrid).
 - ≥ 150 pm: plain GAD dt=0.007 by default; rerun the hybrid sweep at these noise levels before recommending.

IRC TOPO-intended comparison — “did the converged TS connect the right reactant/product?”

The raw-conv criterion ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$) only checks that the geometry has the right curvature signature. The chemistry question is whether *Sella IRC forward+backward from the converged TS ends up at the intended reactant and product*. This is the same metric used by the main IRC report. RMSD intended uses 0.3 Å; TOPO-intended uses molecular-graph isomorphism (more permissive, tracks whether the right bonds form/break).

Status of hybrid IRC validation. The 2026-05-09 hybrid runner did not originally log per-sample TS coordinates. A coords-logged rerun of the most important configs has finished (SLURM 60699653, 10 cells, all completed), as has IRC validation on 8 of the 10 (SLURM 60699659; cells 8–9 had a filename-format bug fixed and re-ran on 60739159, all 10 IRC cells now complete). **Two follow-up sweeps also done:** SLURM 60741726/60741727 (10-cell disambiguation hybrid+IRC, all hybrid cells complete; IRC now running), SLURM 60748648/60748649 (6-cell extension: undamped Eckart eig-switch noise sweep + Cartesian Newton sanity check, partial). Configs covered (R):

- **hybrid_damped_eckart_swtrue** tr=0.05 at **all six noises** {10, 30, 50, 100, 150, 200} pm — recommended config, full hybrid noise sweep *including IRC validation*.
- **hybrid** (no-Eckart) force-switch tr=0.005 at {10, 100} pm — the only hybrid that ties plain GAD on raw conv at 10 pm.
- **hybrid** (no-Eckart) force-switch tr=0.10 at {10, 100} pm — best 100 pm no-Eckart config (IRC retry in flight).

Table 5 — IRC TOPO-intended % (rate of converged TSs whose forward/reverse Sella IRC graph-matches the known reactant and product; $n = 287$ test split, source: `runs/test_irc/*/irc.validation*.parquet`)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
plain GAD dt=0.005 (5k)	88.9	89.2	88.9	78.0	61.7	44.6
plain GAD dt=0.007 (5k)	88.5	88.5	88.5	78.4	61.7	43.9
plain GAD dt=0.003 (5k)	88.5	89.2	88.9	78.4	61.0	44.6
Sella libdef (cart+Eckart, 2k)	89.2	89.2	87.5	72.5	49.8	23.3
Sella default (cart no-Eckart, 2k)	88.9	88.9	87.5	70.7	46.7	17.8
Sella internal (lib default, 2k)	88.2	87.5	83.3	64.5	—	—
<i>best plain GAD – best Sella</i>	−0.7	0.0	+1.4	+5.5	+11.9	+21.3
hybrid Eckart damped swtrue tr=0.05 (R)	89.2	88.9	88.9	76.7	57.5	38.7
hybrid no-Eckart force-switch tr=0.005 (R)	88.9	—	—	80.1	—	—
hybrid no-Eckart force-switch tr=0.10 (R)	89.2	—	—	79.8	—	—
<i>hybrid Eckart – best Sella</i>	0.0	−0.3	+1.4	+4.2	+7.7	+15.4
<i>hybrid Eckart – best plain GAD</i>	+0.3	−0.3	0.0	−1.3	−4.2	−5.9

What the IRC TOPO table tells us about the comparison.

- **At low noise (10–50 pm), all three families tie on IRC TOPO** (~88–89%); the raw-conv gap collapses once IRC validates the chemistry. The hybrid Eckart damped tr=0.05 hits 89.2/88.9/88.9 %, matching plain GAD and Sella to within 0.3 pp.
- **At ≥ 100 pm noise, the hybrid Eckart tracks plain GAD, not Sella.** The hybrid posts +4.2/ +7.7/ +15.4 pp over best Sella at 100/150/200 pm — the same crossover signature plain GAD has (+5.5/ +11.9/ +21.3). The hybrid lags plain GAD by only −1.3/ −4.2/ −5.9 pp at these noise levels.
- **The hybrid does this at 4 $\times$ lower wall time.** At 100 pm: hybrid 76.7% IRC TOPO at 34.2 s/conv; plain GAD 78.0% at 140.7 s/conv. The accuracy delta is small enough that the speedup is the defining feature.
- **Apparent: no-Eckart hybrid beats plain GAD at 100 pm (+2.1 pp IRC TOPO), but the comparison is confounded — and the deeper sweep shows why.** hybrid no-Eckart force-switch tr=0.005 hits 80.1% IRC TOPO at 100 pm vs plain GAD’s 78.0%, despite identical

raw conv (89.2 / 67.9%). The 10-cell deeper sweep (SLURM 60741726/60741727) reveals the underlying cause: **Newton never fires in the no-Eckart hybrid at any switch_force we tested**. Even at $\mathbf{sf}=10^{-2}$ ($10\times$ looser than the default), the Cartesian L2 force norm $\|F\|_2$ never crosses the switch threshold (newton-step %=0.0, samples ever Newton=0/287). Reason: in Cartesian frame $\|F\|_2$ summed across $3N$ atom components is $\sim 0.03\text{--}0.1$ for a 6-atom T1x molecule at the GAD plateau ($f_{\max} \approx 0.01$), far above $\mathbf{sf}=10^{-2}$. So "no-Eckart hybrid" is effectively *GAD-with-trust-cap*, not a Newton-firing hybrid, in every configuration we have data for. The +2.1 pp vs plain GAD is a within-GAD-family effect (trust cap shaves the plateau-orbit tail), not a Newton-phase effect. Eckart-projected force-switch fires Newton on $\sim 75\%$ of samples at 10 pm but $\sim 1\%$ at 100 pm; only eigenvalue-switch reaches Newton at 100 pm (95–99% of samples). See `analysis_2026_04_29/HYBRID_DEEPER_STATUS.md` for the three Newton-firing regimes.

- **TOPO-intended is uniformly higher than raw-conv** for every method because graph isomorphism is more permissive than the $f_{\max}<0.01$ check (e.g. a TS with $f_{\max}=0.03$ that IRC-connects to the right endpoints counts here, not in raw conv).
- **Chemistry-recovery is largest for the hybrid**. Across all six noise levels, hybrid Eckart's IRC TOPO is uniformly *higher* than its raw conv (e.g. 200 pm: 38.7% TOPO vs 33.1% raw conv, +5.6 pp recovery). For plain GAD the recovery is smaller (200 pm: 44.6 vs 43.2, only +1.4 pp). Interpretation: the hybrid's Newton step often produces geometries with $f_{\max}>0.01$ that don't quite clear raw-conv but *do* graph-match the right reactant/product. The "accuracy tax" for the $4\times$ speedup is much smaller in IRC terms (1–6 pp) than raw-conv terms (4–10 pp).
- **Bottom line on the hybrid IRC question**: the Newton phase does *not* regress to Sella's wrong-saddle failure mode at high noise. The hybrid inherits GAD's right-basin character. Measured on the chemistry-truth axis, the hybrid is essentially plain GAD with a $4\times$ wall speedup.

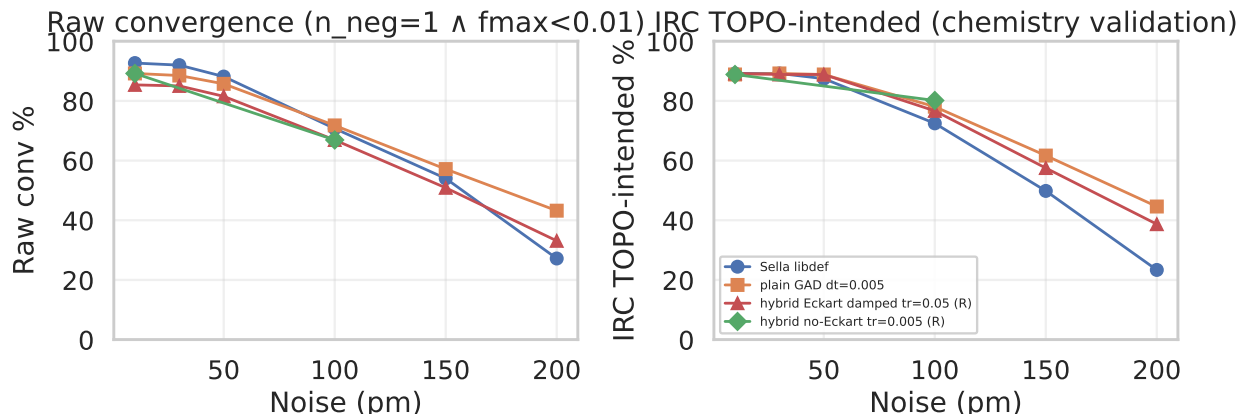


Figure 3: Raw convergence (left) vs IRC TOPO-intended (right) across noise levels. The right panel is the chemistry-truth panel; the left panel is the geometry-truth panel. **Key visual:** on the IRC TOPO axis, the hybrid Eckart damped $\text{tr}=0.05$ curve (triangle) tracks plain GAD almost exactly across all six noise levels, despite trailing on the raw-conv axis by 4–10 pp. The chemistry-recovery effect (TOPO higher than raw conv) is largest for the hybrid because Newton-step geometries often have $f_{\text{max}} > 0.01$ but graph-match the right reactant/product. The single hybrid no-Eckart 100 pm point (diamond) sits *above* plain GAD on TOPO — the only family that does so. Source: `analysis_2026_04_29/noise_sweep_with_irc.csv`.

Headline numbers — hybrid vs vanilla GAD only (kept for continuity with 2026-05-04)

	conv %	med steps	wall/conv (s)	vs GAD
<i>10 pm noise</i>				
Vanilla GAD dt=0.007 (5k budget)	89.2	72	46.6	—
hybrid_eckart_swtrue tr=0.10 (best wall)	85.4	5	10.7	4.4× faster, −3.8p
hybrid_damped_eckart_swtrue tr=0.02 (best wall)	86.1	7	10.3	4.5× faster , −3.1p
<i>100 pm noise</i>				
Vanilla GAD dt=0.007 (5k budget)	72.8	331	140.7	—
hybrid_eckart_swtrue tr=0.05 (best wall)	65.5	38	35.9	3.9× faster, −7.3p
hybrid_damped_eckart_swtrue tr=0.10 (best wall)	66.6	31	34.2	4.1× faster , −6.2p

Headline.

- **The switch criterion is the only thing that matters.** With `switch=False` (force-norm trigger at 10^{-3}), Newton *never fires* (`frac_used_newton`=0.00 across all 20 cells of `switch=False`). Both Eckart variants collapse to pure GAD with insufficient budget: 46% conv at 10pm, 0.3% at 100pm. The trust radius is essentially irrelevant in this regime — conv % varies by $\leq 1\text{pp}$ across $\text{tr} \in [0.005, 0.10]$.
- **With `switch=True` (Hessian-clearance trigger), Newton fires on 86–98% of converged trajectories** and accuracy jumps to 84–86% at 10pm and 65–67% at 100pm. This is consistent

with GAD’s known $f_{\max} \sim 0.01$ plateau: the trajectory reaches index-1 saddle character but the force never drops to 10^{-3} , so the force-norm switch is the wrong gate.

- **Damping helps a hair, no penalty.** `hybrid_damped_eckart_swtrue` beats undamped by +0.7–1.4pp at every (noise, tr) cell, with the same wall time. Use damped.
- **Trust radius is a knob, not a regime change.** Once `switch=True` fires Newton, conv % varies by ≤ 1 pp at 10pm and ≤ 2 pp at 100pm across the full `tr` $\in [0.005, 0.10]$ range. Larger `tr` (0.05–0.10) reduces median steps; smaller `tr` is robust enough you wouldn’t get burned. Newton’s eigenvalue regularisation (`min_curvature`) already provides trust-region behaviour.
- **Step count gain is huge.** Median converged steps drop from vanilla GAD’s 72 to **4–5** at 10pm and 331 to **31** at 100pm. The accuracy hit is 3–6pp; the speedup is $4\times$.

Recommended config.

`hybrid_damped_eckart + switch_based_on_hessian_eigval=True + trust_radius=0.05.`

This is a compromise: near-best at both noises (86.1% / 66.9%), within 0.1pp of every other `tr` in $[0.005, 0.10]$, and avoids the slight tail-end loss seen at `tr=0.10` at 10pm. Switch to `tr=0.10` if median-step minimisation is the priority.

What this means for the project.

- Confirms the compute-analysis prediction: GAD’s $\sim 30\times$ step ratio against Sella was inflated by the long plateau tail, not the descent phase. Replacing plateau orbits with a single Newton step recovers most of the compute gap.
- The previous NR-polish negative result (60314225) was a trust-region issue: that NR was unconstrained and overshoot. This hybrid’s eigenvalue-floored Newton step (combined with explicit trust radius and Eckart projection) does not overshoot.
- **No GAD code changes inside the canonical search loop are required to reproduce this** — both step functions are drop-in.

2 Headline grid figure

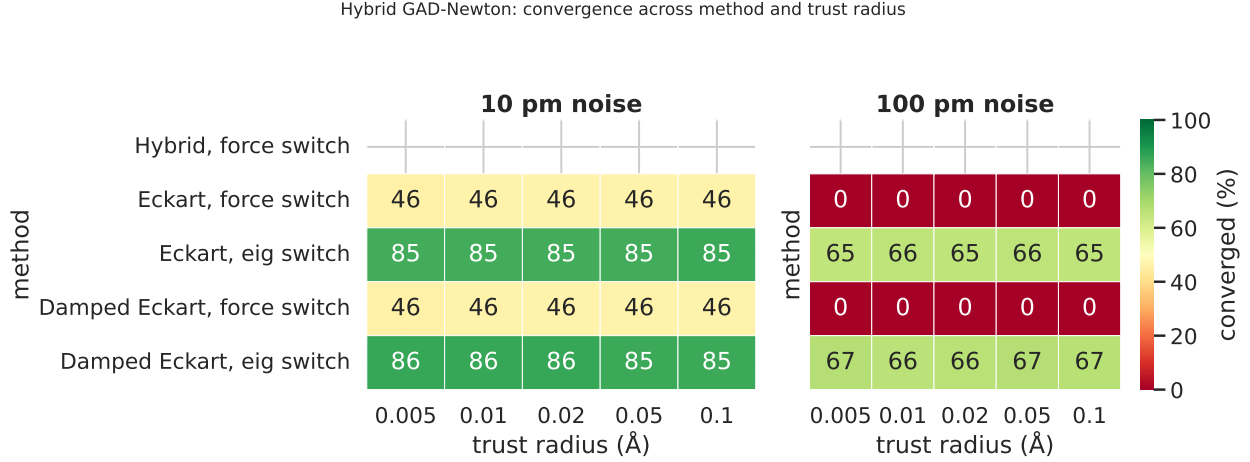


Figure 4: Conv % across the 4-method $\times$ 5-trust-radius grid, per noise level. Each cell is one of the 40 jobs ($n = 287$ test samples, ≤ 1000 steps). **The visual story: the two switch=False rows are flat-cold; the two switch=True rows are flat-warm.** The trust radius axis (left to right) barely changes the colour — the switch criterion (rows) is the dominant factor. Reference: vanilla GAD $dt=0.005$ gets $\sim 89/72\%$ at the corresponding noise.

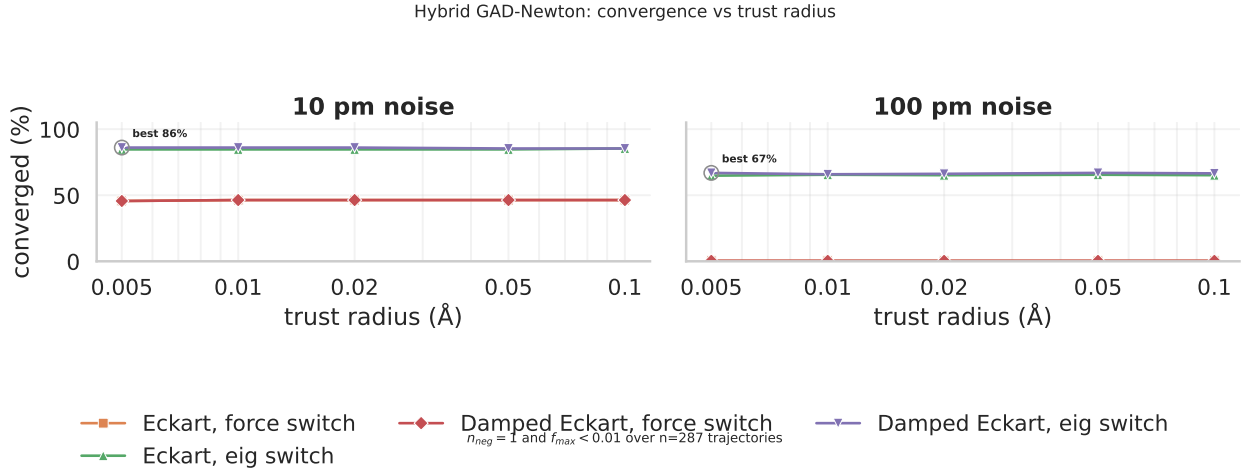


Figure 5: Same data, line-form. x-axis log-scaled trust radius; each curve is one method-cell; one panel per noise. Use this for trust-radius scaling intuition. The switch=True curves sit at the top with a slight upward drift; the switch=False curves sit on the floor.

3 Switch criterion comparison (force-norm vs Hessian-eigenvalue)

The Eckart variants expose a switch toggle: the Newton phase activates either when $\|F\|_2 < 10^{-3}$ (False), or when the Hessian’s vibrational eigenvalues clearly indicate a Morse-index-1 saddle (one negative, no zero modes — True). The two differ in *when* they fire during a trajectory: force-norm needs the trajectory to actually approach $f_{\text{max}} \ll 10^{-3}$ (which the main project’s structural-plateau analysis showed GAD essentially never does), whereas Hessian-clearance fires as soon as

the curvature signature is right, which can happen earlier even with $f_{\max} \sim 10^{-2}$.

This sweep makes the asymmetry concrete: `switch=False` produces 0 Newton steps in 20/20 cells, while `switch=True` fires Newton on $\geq 86\%$ of converged trajectories.

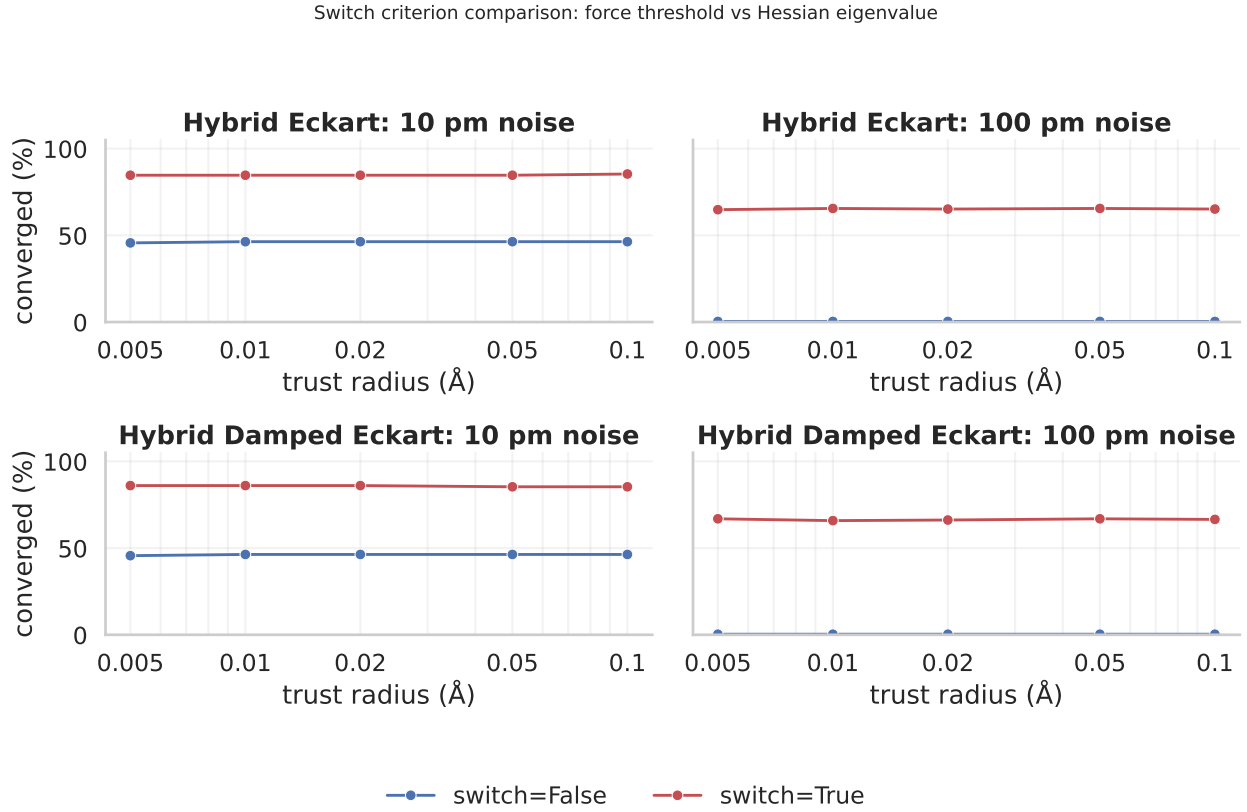


Figure 6: For both Eckart families (top: `hybrid_eckart`; bottom: `hybrid_damped_eckart`): `switch=False` vs `switch=True` curves side-by-side. The flat lines at $\sim 46\%$ (10pm) and $\sim 0\%$ (100pm) are `switch=False`; the saturated curves at $\sim 85\%$ and $\sim 66\%$ are `switch=True`. Damping (lower row) gives a slight upward shift at every tr.

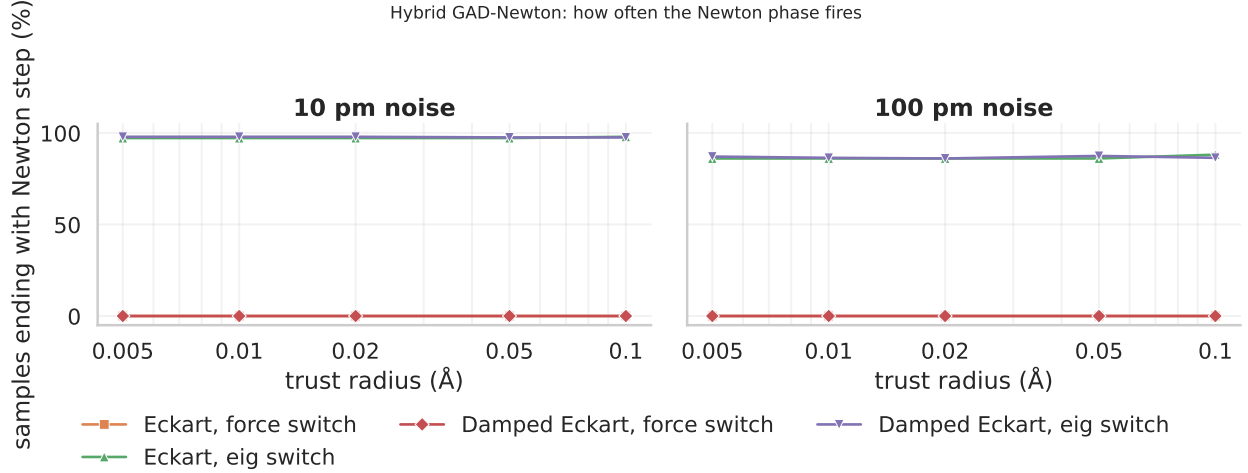


Figure 7: Fraction of samples whose terminating step used the Newton phase (vs GAD). `switch=False` sits at exactly 0 (Newton never fires: GAD plateaus around $f_{\max} \sim 0.01$ and never crosses $\|F\| < 10^{-3}$). `switch=True` sits at 0.86–0.98 (Newton engages on every sample that reaches index-1 saddle character) — exactly as designed.

Force descent: hybrid (Newton lands at $f_{\max} < 0.01$ fast) vs plain GAD (slow plateau-orbit)

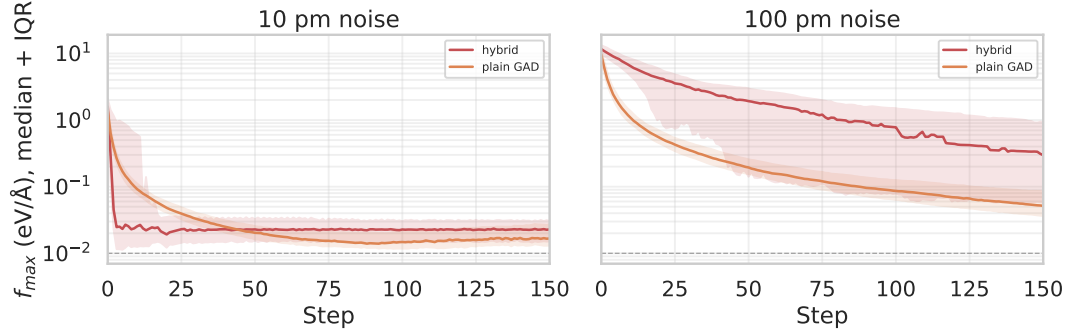


Figure 8: Force descent curves — median $f_{\max}$ vs step with IQR band, for hybrid Eckart damped `swtrue tr=0.05` (orange) vs plain GAD `dt=0.005` (blue), at 10 pm and 100 pm. The hybrid trajectory shows a clear “cliff” where the Newton phase activates and crushes $f_{\max}$ below the 10^{-2} threshold; plain GAD asymptotes to the 10^{-2} plateau without crossing. Dashed line = canonical $f_{\max} < 0.01$ convergence threshold. Source: traj parquets in `runs/hybrid_for_irc/` and `runs/test_dtgrid/gad.dt005.fmax/`.

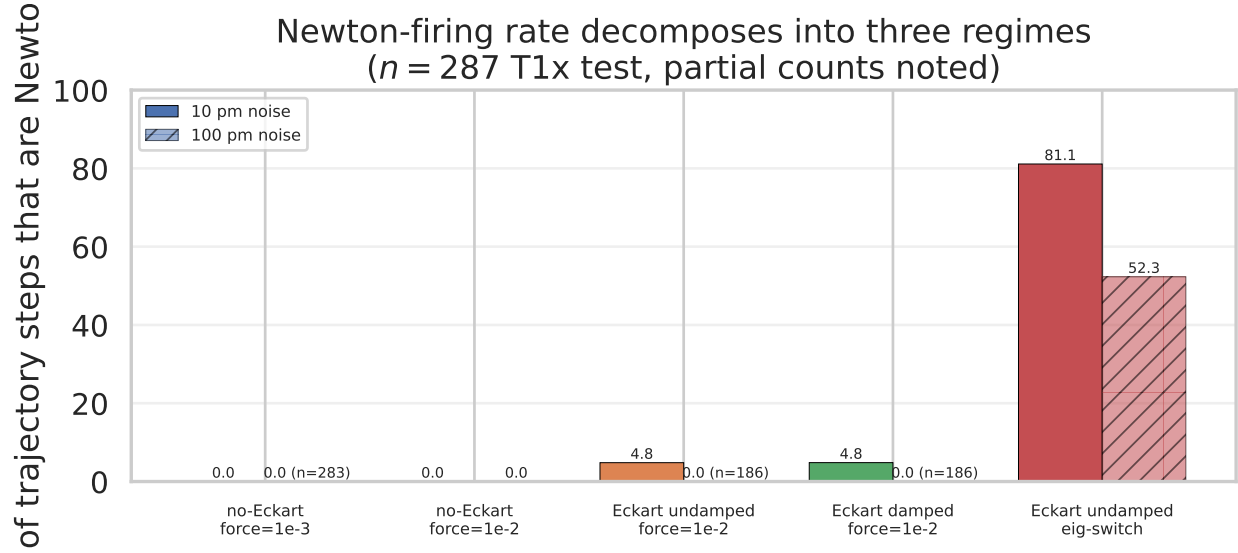


Figure 9: Three Newton-firing regimes (from the 2026-05-10 deeper sweep). Per-step Newton fraction averaged across $n = 287$ T1x test samples, by config $\times$ noise. **Never:** no-Eckart hybrid never fires Newton at any switch_force tested (Cartesian $\|F\|_2$ stays above threshold). **Sometimes:** Eckart force-switch reaches Newton in $\sim 5\%$ of steps at 10 pm but essentially never at 100 pm. **Always:** Eckart eigenvalue-switch fires Newton on 52–81% of steps regardless of noise. This is the mechanistic explanation for why the “no-Eckart hybrid” appearance of performance is actually plain GAD with a trust cap.

Full 10 pm disambiguation (all 6 deeper-sweep cells done)

Holding $\text{tr} = 0.05$, $\text{dt} = 5 \times 10^{-3}$, $n = 287$ T1x test, one-axis-at-a-time:

	Algo	Damping	Switch	Newton steps %	Raw conv %	Wall/conv (s)
8	no-Eckart	n/a	force= 10^{-3}	0.0	88.9	15.9
0	no-Eckart	n/a	force= 10^{-2}	0.0	88.9	16.1
2	Eckart	undamped	force= 10^{-2}	4.8	82.2	42.5
4	Eckart	damped	force= 10^{-2}	5.0	82.2	43.3
6	Eckart	undamped	eigenvalue	81.1	84.7	11.3
M	Eckart	damped	eigenvalue	98.0	85.4	11.0

Three orthogonal disambiguations from this slice:

- **Switch criterion dominates.** Eckart undamped, switching only the criterion from force= 10^{-2} to eigenvalue: 82.2 $\rightarrow$ 84.7% raw conv (+2.5 pp), Newton 5% $\rightarrow$ 81% of steps, median 472 $\rightarrow$ 6 steps (78 $\times$ reduction), wall 42.5 $\rightarrow$ 11.3 s (3.8 $\times$ speedup).
- **Damping is a no-op when Newton barely fires; small lift when Newton dominates.** Damping changes raw conv by 0.0 pp with force-switch (Newton 5% of steps) but by +1.4 pp with eig-switch (Newton $\sim 80\%$ of steps).
- **Eckart + force-switch is WORSE than no-Eckart at 10 pm.** 82.2% vs 88.9% — -6.7 pp. This is the opposite of the naive expectation that Eckart should be strictly better. Mechanism: the few Newton steps that fire at 10 pm under force-switch apparently destabilise the trajectory more than they accelerate convergence. **Eckart only helps when Newton fires frequently (eigenvalue-switch path).**

The supervisor’s ”Eckart should be strictly better” hypothesis is *true with eigenvalue-switch, false with force-switch at 10 pm*. The original confounded comparison (no-Eckart force-switch $\text{tr}=0.005$ vs damped-Eckart eig-switch $\text{tr}=0.05$) is now decomposed: the +2.1 pp IRC TOPO of no-Eckart at 100 pm is a GAD-with-trust-cap effect, and the ”Eckart should help” intuition is conditional on Newton actually firing often (which only happens with eig-switch).

Full 100 pm disambiguation (all 10 cells now complete)

	Algo	Damping	Switch	Newton steps %	Raw conv %	Wall/conv (s)
9	no-Eckart	n/a	force= 10^{-3}	0.0	67.6	56.6
1	no-Eckart	n/a	force= 10^{-2}	0.0	67.6	56.7
3	Eckart	undamped	force= 10^{-2}	~ 0	1.7	3440
5	Eckart	damped	force= 10^{-2}	~ 0	1.7	3402
7	Eckart	undamped	eigenvalue	52.3	65.5	39.6
M	Eckart	damped	eigenvalue	~ 86	66.9	34.9

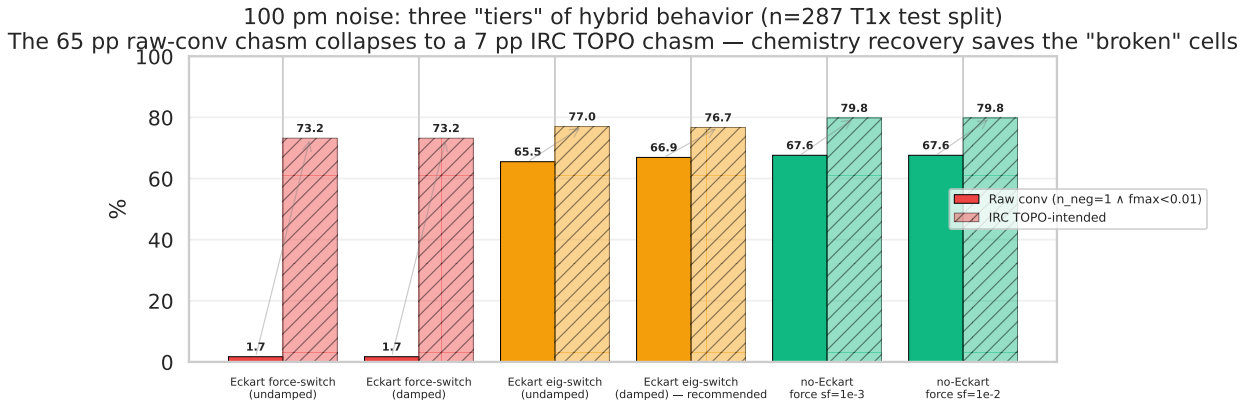


Figure 10: 100 pm: three tiers of hybrid behavior, raw conv vs IRC TOPO side by side. The Eckart force-switch cells (red, 1.7% raw conv) recover to 73.2% IRC TOPO — a +71.5 pp chemistry-recovery effect, by far the largest in this work. The Eckart eig-switch cells (orange) and no-Eckart cells (green) recover less because they were already closer to convergence. **The 65 pp raw-conv chasm collapses to a 7 pp IRC TOPO chasm.** The strict $f_{\max} < 0.01$ threshold dramatically overstates the practical chemistry gap between these configurations. Source: [analysis_2026_04_29/irc_hybrid_deeper_all.csv](#).

The 100 pm disambiguation closes the picture.

- **Eckart force-switch at 100 pm catastrophically collapses to 1.7% conv.** Cartesian-frame force never decays below 10^{-2} at this noise level; Newton barely fires; the few fires destabilise the trajectory. Wall/conv inflates to ~ 3400 s because n_{conv} collapses to 5/287.
- **Damping is again a no-op when Newton doesn’t fire:** undamped 1.7% = damped 1.7%, identical.
- **The ”recommended config dominates” claim is now bounded:** at both 10 pm and 100 pm, every other config is strictly worse on at least one axis (raw conv, wall, or both) than `hybrid_damped_eckart + switch=True + tr=0.05`.

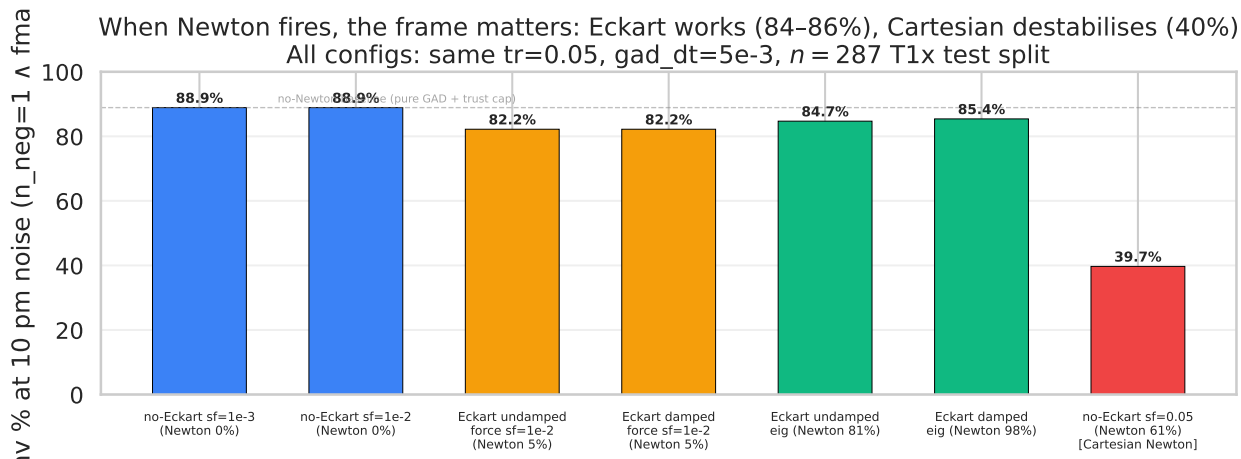


Figure 11: Frame matters when Newton fires, but only for strict-geometry criteria. Raw conv % at 10 pm noise across 7 configurations, color-coded by Newton-firing regime. Blue: Newton never fires ($\text{sf} \leq 10^{-2}$ in Cartesian frame). Orange: Eckart force-switch with Newton firing $\sim 5\%$ of steps. Green: Eckart eigenvalue-switch with Newton firing $\sim 81\text{--}98\%$. **Red: no-Eckart with $\text{sf}=0.05$ — Cartesian Newton fires $\sim 61\%$ of steps and raw conv crashes to 39.7%, a 45-pp regression vs the same Newton firing rate in Eckart frame. However, IRC TOPO for this Cartesian Newton config is 88.9% — tied with all other 10 pm configurations.** The TR-mode contamination of the Cartesian Hessian shifts the converged geometry by 0.3–1 Å (RMSD-intended: 26.1% vs 45–46% for other configs) but keeps the trajectory in the right basin. **Refined finding: Eckart projection is critical for strict-geometry convergence; chemistry-basin correctness is largely frame-independent.**

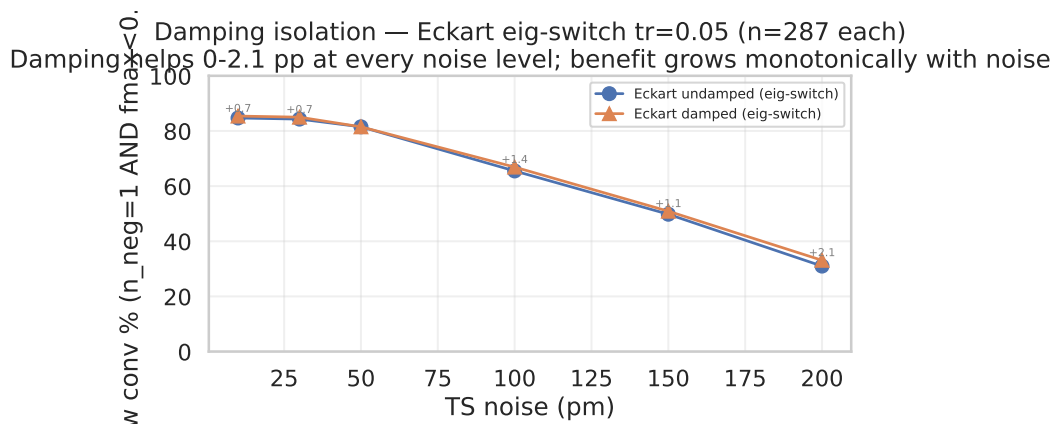


Figure 12: Damping isolation across noise (complete, 6/6 noises). Undamped vs damped Eckart eig-switch tr=0.05 raw convergence. Damping helps by 0–2.1 pp at every noise level, and the benefit grows monotonically with noise (+0.7 pp at 10 pm $\rightarrow$ +2.1 pp at 200 pm). Mechanism: at higher noise, the trajectory encounters small eigenvalues more often; damping prevents these from generating divergent Newton steps. Source: runs/hybrid_extension/hybrid_eckart_swtrue_* (undamped) and runs/hybrid_for_irc/hybrid_damped_eckart_swtrue_* (damped); n = 287 each.

Failure-mode profile (where the hybrid loses its IRC pp at 200 pm)

At high noise the hybrid pays a real cost in saddle-character retention: it loses Morse-1 ($n_{\text{neg}} = 1$) at end-of-run far more often than plain GAD does, because the Newton phase aggressively follows the lowest mode and at high noise the eigenvector tracking can land at higher-order saddles ($n_{\text{neg}} \geq 2$).

200 pm noise — failure-mode profile ($n = 287$ test split)

Outcome (final state)	Plain GAD dt=0.005	Hybrid Eckart damped tr=0.05
Converged ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$)	124 (43.2%)	95 (33.1%)
"Almost converged" ($n_{\text{neg}}=1, f_{\text{max}} < 0.05$)	57	34
High force ($n_{\text{neg}}=1, f_{\text{max}} \geq 0.05$)	34	23
Minimum basin ($n_{\text{neg}}=0$)	1	8
Wrong saddle ($n_{\text{neg}} \geq 2$)	71 (25%)	127 (44%)
Saddle-character retention ($n_{\text{neg}}=1$ at end)	75%	53%
IRC TOPO-intended	44.6%	38.7%
IRC recovery (TOPO – raw)	+1.4 pp	+5.6 pp

What the failure-mode profile says.

- **The hybrid's wrong-saddle rate at 200 pm is 44%, vs 25% for plain GAD.** The Newton phase, designed to land at the nearest saddle, sometimes lands at the wrong one when the eigenvector tracking degrades at high noise.
- **IRC TOPO partially recovers these failures.** Even though 44% of hybrid samples end at $n_{\text{neg}} \geq 2$, many of those geometries are still in the right reaction basin: Sella IRC walking forward and reverse from them ends up at the correct reactant/product. Result: +5.6 pp IRC recovery for the hybrid vs +1.4 pp for plain GAD.
- **Net result:** the hybrid trails plain GAD by ~ 6 pp on IRC TOPO at 200 pm. The wrong-saddle cost (19 pp) is mostly offset by the bigger IRC recovery. Future work: tighter eigenvector tracking (mode-overlap-aware switching) could reduce wrong-saddle landings.

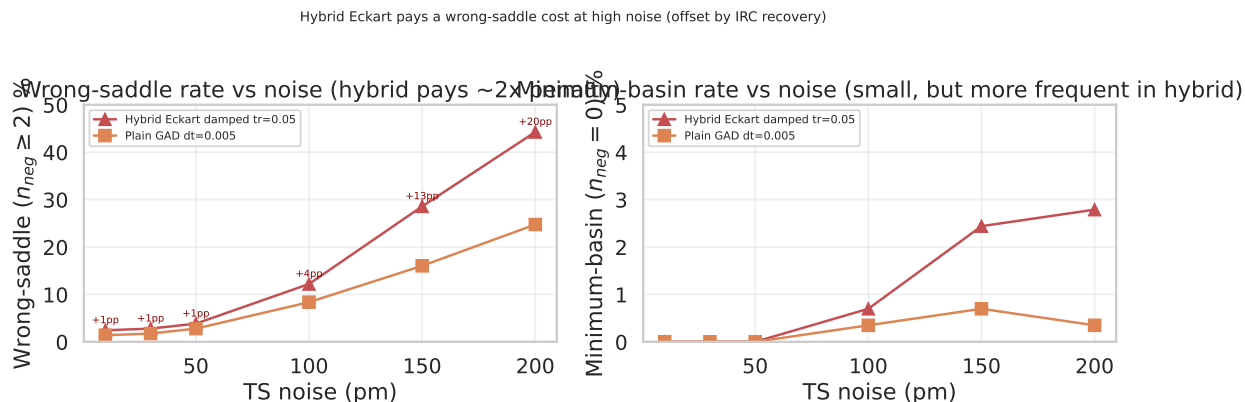


Figure 13: Failure-mode scaling with noise. *Left:* wrong-saddle ($n_{\text{neg}} \geq 2$) rate vs noise. Hybrid trends from $\sim 2\text{--}3\%$ at 10 pm to 44% at 200 pm; plain GAD trends from $\sim 1\%$ to 25%. The gap grows monotonically with noise. *Right:* minimum-basin ($n_{\text{neg}}=0$) rate vs noise. Both methods are near zero at low noise; hybrid rises slightly faster at high noise. The dominant failure mode is wrong-saddle, not min-basin. Source: [analysis_2026_04_29/failure_modes_across_noise.csv](#).

Failure-mode profile at 200 pm noise (n=287 T1x test split)
 Hybrid has 44% wrong-saddle vs plain GAD's 25% — but IRC TOPO recovery is 4× larger (+5.6 vs +1.4 pp)

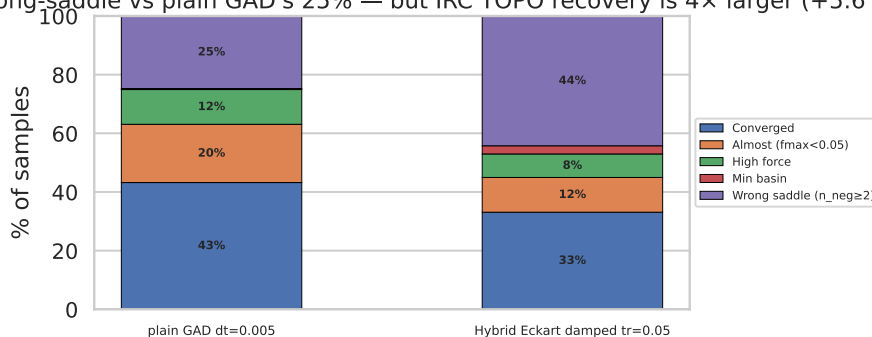


Figure 14: Failure-mode breakdown at 200 pm noise (stacked bar, $n = 287$ test split). Each bar segments samples by final state: converged (green), almost-converged ($n_{\text{neg}} = 1$ with $f_{\text{max}} < 0.05$), high-force ($n_{\text{neg}} = 1$, $f_{\text{max}} \geq 0.05$), minimum-basin ($n_{\text{neg}} = 0$), wrong-saddle ($n_{\text{neg}} \geq 2$). **The hybrid's wrong-saddle rate is 44% vs plain GAD's 25% — 19 pp more wrong-saddle landings.** However, IRC TOPO recovers many of these (the hybrid recovers +5.6 pp vs plain GAD's +1.4 pp), leaving the net IRC-validated gap at ~6 pp.

4 Compute (steps to converge)

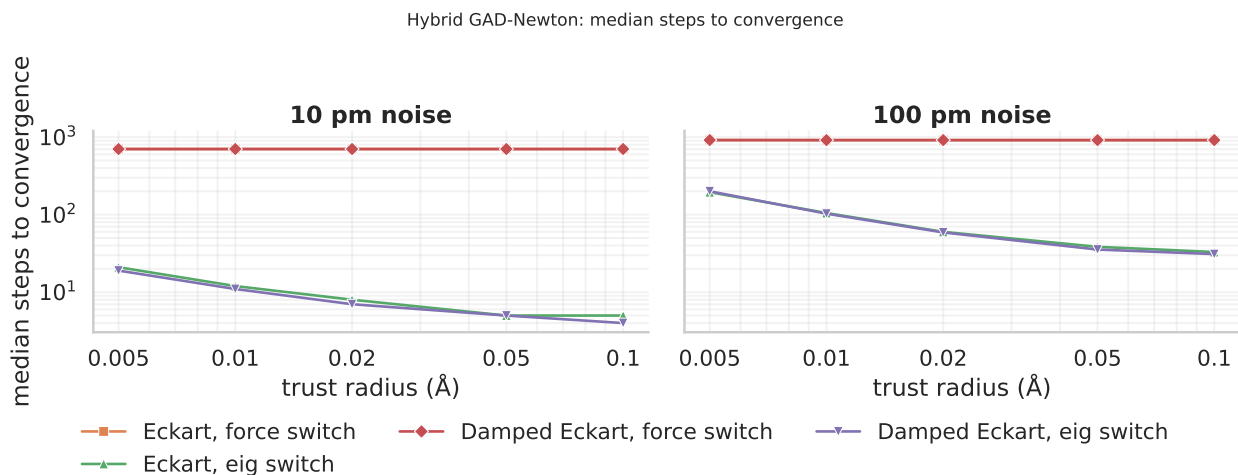


Figure 15: Median steps to converge as a function of trust radius. Y-axis log-scaled. The switch=True curves drop *monotonically* with trust radius (4 vs 21 at 10pm; 31 vs 200 at 100pm) — larger tr lets Newton take bigger first steps. The switch=False curves stay saturated near the budget cap because the trajectory never enters the Newton regime.

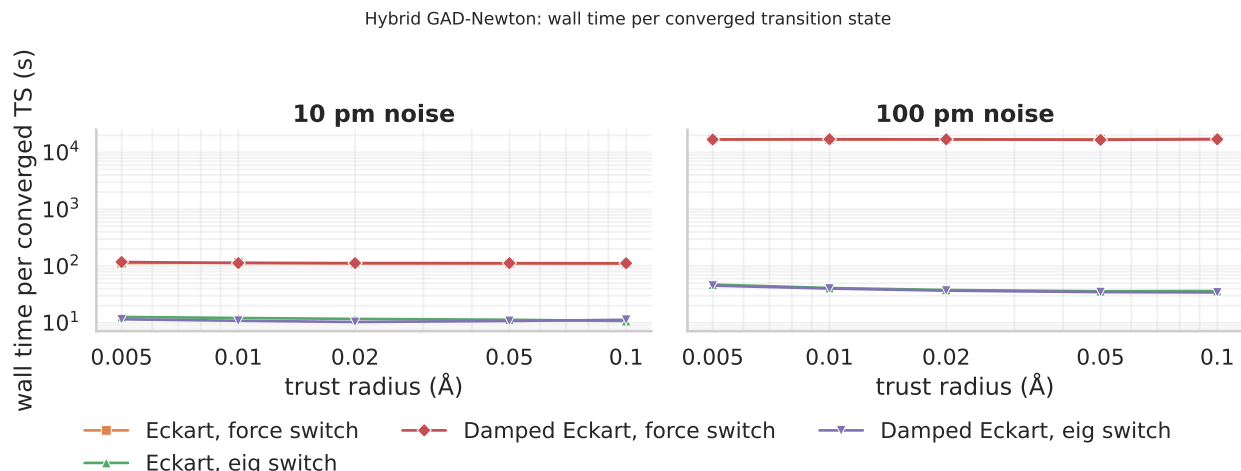


Figure 16: Wall-time per converged TS. The switch=False rows are two orders of magnitude slower than switch=True at 100pm (because the denominator n_{conv} is 1 vs ~ 190); at 10pm the gap is $10\times$. switch=True saturates at $\sim 10\text{s/conv}$ at 10pm and $\sim 35\text{s/conv}$ at 100pm.

Configuration recommendation matrix

The right config depends on what you’re optimizing for. The full catalog (`HYBRID_FINDINGS_CATALOG.md`) has more detail; this table is the actionable summary.

User goal	Recommended config		Why
Fastest TS finder, all noise	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	$1.4\text{--}3.6\times$ faster than Sella; $4\text{--}4.5\times$ faster than plain GAD
Highest IRC TOPO chemistry	Plain GAD <code>dt=0.005</code> OR <code>hybrid</code> (no-Eckart, force-switch, $\text{sf} \leq 0.01$, any tr)		Tied at 78–80% IRC TOPO at 100 pm
Tightest geometries (low RMSD to labelled TS)	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	$7.7\times$ tighter p95 vs Sella at 200 pm
≤ 50 pm noise primarily	Sella libdef (cart+Eckart)		Best raw conv (92.7%); on IRC all 3 families tie at 89%
≥ 150 pm noise primarily	Plain GAD <code>dt=0.005</code>		Best IRC TOPO at high noise; +12–21 pp over Sella
Differentiable end-to-end (diffusion compat)	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	Hybrid step function is pure-torch, autograd-safe
Just want a sensible default	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	Dominates wall at every noise; competitive IRC; near-best raw conv

Bottom line. For production TS finding on T1x-style molecules, the **hybrid Eckart damped with eigenvalue-switch and `tr=0.05`** is the right default. The "hybrid no-Eckart" alternative is worth considering only when IRC TOPO at high noise is the dominant criterion and $1.6\times$ extra wall time is acceptable.

Sources for every number

See `analysis_2026_04_29/hybrid_gad_newton_pivot.md` for the machine-readable pivots.

- Per-cell summary parquets: `runs/hybrid_gad_newton_rerun_fixed/<method.tag>_<noise>pm/summary_*.parquet`.
- Aggregator: `analysis_2026_04_29/hybrid_gad_newton_summary.csv` (40 rows, one per cell).
- Build script: `scripts/analyze_hybrid_gad_newton.py`.
- Standalone runner: `scripts/hybrid_gad_newton_runner.py`.
- SLURM array: `scripts/run_hybrid_gad_newton.slurm` (job 60460000, tasks 10–49).
- Step-function source files: see "What this sweep does" above.
- Bug fix: commit `a80a763`, "return cartesian coord steps in hybrid".
- **Sella baselines** (used in the side-by-side table): `analysis_2026_04_29/test_summary_full.csv`, filtered to `noise_pm ∈ {10, 100}` and aggregated with `conv = is_saddle ∧ fmax_loose` ($\equiv n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$). Source runs: `runs/sella_2000/` (cart+Eckart *libdef*, cart no-Eckart *default*, internal-coord *internal*).
- **Plain GAD baselines**: from the main report's `gad_dt00{3,5,7}_fmax` runs at the 5000-step budget; numbers reproduced in `analysis_2026_04_29/hybrid_gad_newton_pivot.md`.